

Model Compression for Very Short-Term Solar Irradiance Forecasting Using Infrared Sky Images

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Abstract

Accurate short-term solar irradiance forecasting is essential for reliable photovoltaic power integration. While convolutional neural networks (CNNs) have demonstrated strong performance for sky-image-based forecasting, deploying deep learning models on resource-constrained edge devices remains challenging due to their computational and memory requirements. However, the impact of model compression on solar forecasting accuracy and deployment efficiency has not been systematically investigated. In this work, we present a comprehensive benchmark of structured and unstructured pruning for infrared sky image-based solar irradiance forecasting using infrared sky images from the Girasol dataset. We evaluate four widely used CNN architectures ResNet-18, EfficientNet-B0, MobileNetV2, and MobileNetV3-Small along with a lightweight custom baseline model (TinyCNN) across multiple pruning levels ranging from 20% to 80%. Models are assessed using forecasting accuracy, model size, parameter reduction, and inference latency. Experimental results show that MobileNetV2 achieves the strongest baseline forecasting performance, with an RMSE of 37.11 W/m². Structured pruning yields substantial parameter and latency reductions but is architecture-dependent, with some models degrading sharply even at the lowest pruning level tested. Unstructured pruning preserves accuracy more consistently across architectures and substantially reduces compressed checkpoint size, but provides little to no inference speed improvement on conventional hardware. TinyCNN tolerates the highest compression of any architecture under both strategies, though from a much higher baseline error. Overall, the two strategies offer complementary trade-offs structured pruning favoring latency and parameter reduction, unstructured pruning favoring accuracy preservation demonstrating that meaningful compression is achievable for CNN-based solar irradiance forecasting, with implications for edge deployment.

Keywords: Solar irradiance forecasting, infrared sky imaging, convolutional neural networks, model pruning, edge deployment

1 Introduction

The increasing adoption of photovoltaic energy systems has made accurate short-term solar irradiance forecasting essential for grid stability, energy management, and power dispatch [Barhmi et al., 2026, Alnajadah et al., 2025]. Since rapid cloud movement can cause large intra-hour fluctuations in solar power generation, forecasting Global Solar Irradiance (GSI) remains an important research problem [Nie et al., 2024]. Ground-based sky imaging systems are particularly effective for short-term irradiance forecasting because they directly capture local cloud dynamics at high temporal resolution [Diagne et al., 2013, Paletta et al., 2023]. The Girasol dataset introduced by Terrén-Serrano et al. [Terrén-Serrano et al., 2021] provides synchronized infrared sky images and pyranometer measurements for irradiance estimation. Infrared imaging is useful in this context because it captures cloud structure while being less affected by strong solar glare than visible-light imaging.

Recent deep learning approaches have improved image-based irradiance forecasting by learning cloud-related features directly from sky images [Klotz and Nayar, 2026]. CNN-based models, including lightweight architectures such as MobileNetV2, have shown promising performance on the Girasol dataset [Nijhum et al., 2024]. Complementary research has also leveraged RGB and near-infrared (NIR) sky images for solar forecasting, highlighting the value of multi-spectral imaging for capturing cloud dynamics and irradiance variability [Nijhum et al., 2026]. However, practical deployment remains challenging. Solar monitoring stations and embedded IoT devices often operate under limited memory, computation, and energy budgets, while many CNN-based forecasting models are still too costly for efficient edge inference [Menghani, 2023, Paletta et al., 2023]. This creates a trade-off between forecasting accuracy and deployment efficiency. Model compression provides a practical way to reduce this cost. Pruning can remove redundant parameters or channels from neural networks, while quantization reduces numerical precision to lower storage and computational requirements [He and Xiao, 2023, Fang et al., 2023, Wu et al., 2020, Guo, 2018]. Although these techniques are widely studied in general computer vision, their use for sky-image-based solar irradiance forecasting remains limited.

To the best of our knowledge, no prior work has systematically evaluated pruning techniques across multiple CNN architectures for infrared sky image-based solar irradiance forecasting. Existing studies mainly focus on forecasting accuracy, with limited attention to deployment-oriented metrics such as model size and inference latency. In this work¹, we present a systematic compression benchmark for CNN-based solar irradiance forecasting using infrared sky imagery from the Girasol dataset. Five architectures, ResNet18, MobileNetV2, MobileNetV3-Small, EfficientNet-B0, and a custom baseline (Tiny CNN), are evaluated under multiple structured and unstructured pruning levels. The models are compared using validation RMSE, model size, and inference latency.

The main contributions of this work are summarized as follows:

- We evaluate the performance of several popular lightweight CNN architectures for short-term solar irradiance forecasting using infrared sky imagery.
- We perform both structured and unstructured pruning across multiple compression levels to analyze their impact on forecasting performance.
- We provide a comparative evaluation of forecasting accuracy, model size, and inference latency to study the trade-off between compression and deployment efficiency.

2 Methodology

This study investigates the impact of structured and unstructured pruning on convolutional neural networks (CNNs) for infrared sky image-based solar irradiance forecasting. The overall pipeline consists of four stages: dataset preparation, baseline model training, model compression through pruning, and post-pruning fine-tuning and evaluation. Infrared sky images are used as model inputs, while future global horizontal irradiance (GHI) values measured by a pyranometer serve as prediction targets. Multiple lightweight CNN architectures are evaluated, including MobileNetV2, MobileNetV3-small, EfficientNet-B0, ResNet-18, and a custom Tiny CNN baseline. Baseline models are first trained using a unified training configuration. Structured and unstructured pruning are then independently applied at multiple pruning ratios, followed by fine-tuning and evaluation.

2.1 Problem Formulation

The forecasting task is formulated as a supervised image-to-regression problem. Given an infrared sky image captured at time step t , the objective is to predict the future solar irradiance value at time step $t + L$, where L denotes the forecasting horizon.

$$\hat{y}_{t+L} = f(I_t; \theta), \quad (1)$$

¹<https://github.com/ifran-rahman/solar-forecasting-pruning>

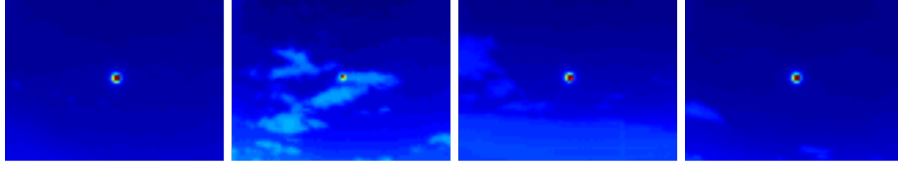


Figure 1: Sample infrared sky images from the Girasol dataset used for solar irradiance forecasting. The images capture a range of sky conditions, including clear, partially cloudy, and overcast scenarios.

where I_t denotes the infrared sky image at time t , $\hat{y}_{t+L}$ denotes the predicted future irradiance value, $f(\cdot)$ represents the CNN forecasting model, and θ denotes the trainable model parameters. The forecasting horizon was fixed at 5 minutes (20 future time steps).

2.2 Dataset Preparation

The Girasol Sky Imaging and Global Solar Irradiance Dataset [Terrén-Serrano et al., 2021] contains synchronized infrared sky images and pyranometer-based irradiance measurements collected at the University of New Mexico for short-term solar forecasting. Figure 1 demonstrate some sample IR images from the dataset. This study uses 59,778 infrared sky images acquired between December 2017 and January 2018. To preserve temporal consistency and prevent data leakage, the dataset was split chronologically, with training samples drawn from 8 December 2017 to 8 January 2018 and validation samples taken from the subsequent 7-day period, consisting of 48,164 raw training images and 11,614 raw validation images. The forecasting horizon was fixed at $L = 20$ future time steps. After removing samples without valid future targets, the final dataset contained 47,564 training samples and 11,474 validation samples. For each sample, the input image at time t was paired with the irradiance measurement at time $t + L$. Images with a resolution of 60×80 pixels were normalized using min-max normalization.

2.3 CNN Architectures

Five CNN architectures were evaluated to analyze the effectiveness of pruning across lightweight and heavy-weight models. These widely adopted architectures are commonly used benchmarks for studying accuracy-efficiency trade-offs. All architectures were adapted for scalar irradiance regression by replacing their original classification heads with a single fully connected regression layer producing one irradiance prediction.

MobileNetV2 is a lightweight CNN architecture based on inverted residual blocks and depthwise separable convolutions, designed for efficient mobile and edge inference. MobileNetV3 Small extends this design by incorporating squeeze-and-excitation modules and hardware-aware neural architecture search for improved computational efficiency. EfficientNet-B0 employs compound scaling of network depth, width, and input resolution to achieve a balanced trade-off between accuracy and computational cost. ResNet18 is a residual CNN architecture that uses skip connections to facilitate stable optimization of deeper networks and serves as the heavyweight reference model in this study. In addition, a custom lightweight architecture, referred to as TinyCNN, was implemented using two convolutional layers with ReLU activations and max-pooling operations, followed by a fully connected regression head. The TinyCNN architecture consists of two convolutional blocks followed by a fully connected regression layer, as summarized in Eq. (2). TinyCNN was designed as an extremely lightweight edge-deployment baseline.

$$\text{Input} \rightarrow \text{Conv}(3 \times 3, 16) \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \rightarrow \text{Conv}(3 \times 3, 32) \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \rightarrow \text{Flatten} \rightarrow \text{FC}(9600 \rightarrow 1) \quad (2)$$

All models were trained from scratch without pretrained weights. For MobileNetV2, MobileNetV3-Small, EfficientNet-B0, and ResNet18, the original classification layers were removed and replaced with regression heads. Adaptive average pooling was additionally applied after feature extraction for the MobileNet-based architectures.

2.4 Structured Pruning

Structured pruning removes entire convolutional filters or channels, reducing parameter count and computational complexity while preserving dense tensor operations compatible with standard hardware [Zhu et al., 2025]. Magnitude-based structured pruning was implemented using the `torch-pruning` framework with L2-norm filter importance:

$$S_i = \|W_i\|_2 = \sqrt{\sum_j w_{ij}^2}, \quad (3)$$

where W_i denotes the weights of filter i . Filters with lower importance scores were pruned first.

Pruning ratios of 20%, 40%, 60%, and 80% were evaluated across all CNN architectures. Following pruning, models were fine-tuned for 3 epochs using Adam with a reduced learning rate of 10^{-4} . Performance was evaluated using RMSE, MAE, parameter reduction, model size, and inference latency.

$$\text{Parameter Reduction (\%)} = \left(1 - \frac{P_{\text{pruned}}}{P_{\text{baseline}}}\right) \times 100, \quad (4)$$

where P_{baseline} and P_{pruned} denote the number of parameters before and after pruning, respectively.

2.5 Unstructured Pruning

Unstructured pruning removes individual weights while preserving the original tensor dimensions, producing sparse weight matrices. L1 unstructured pruning was applied to convolutional and linear layers using the PyTorch pruning utilities. Weight importance was determined using absolute magnitude:

$$S_{ij} = |w_{ij}|, \quad (5)$$

where w_{ij} denotes an individual weight parameter.

Pruning ratios of 20%, 40%, 60%, and 80% were evaluated. After pruning, models were fine-tuned for 3 epochs using Adam with a learning rate of 10^{-4} . Final evaluation included RMSE, MAE, sparsity percentage, model size, and inference latency.

2.6 Training Configuration

All models were trained using the Adam optimizer with a batch size of 256 and a mean squared error (MSE) loss function. To ensure reproducibility, a fixed random seed of 42 was used for NumPy, Python, and PyTorch, with deterministic CUDA operations enabled. The baseline models were trained from random initialization for 50 epochs using an initial learning rate of 0.001. A StepLR scheduler was employed to reduce the learning rate by a factor of 0.5 every 10 epochs, and the checkpoint corresponding to the lowest validation loss was retained for evaluation. For both structured and unstructured pruning experiments, the trained baseline models served as the starting point. Pruning was applied at four pruning levels (20%, 40%, 60%, and 80%), followed by three epochs of fine-tuning using a reduced learning rate of (1×10^{-4}) to recover performance. Structured pruning was implemented using the Torch-Pruning MagnitudePruner with L2-norm importance, while unstructured pruning employed L1-based weight pruning on convolutional and linear layers. In all cases, the best fine-tuned model was selected based on validation loss and used for the final evaluation.

3 Experiments and Results

3.1 Evaluation Metrics

All models were optimized using mean squared error (MSE) loss and evaluated using root mean squared error (RMSE) and mean absolute error (MAE), reported in W/m²:

Table 1: Baseline forecasting performance, parameter counts, and checkpoint sizes for 60×80 infrared sky images. Lower values indicate better forecasting performance.

Model	Params (M)	RMSE (W/m ²)	MAE (W/m ²)	MSE (W ² /m ⁴)	Size (MB)	Latency (ms/img)
TinyCNN	0.015	82.14	61.87	6747.77	0.06	0.0200
MobileNetV2	2.225	37.11	27.14	1376.78	8.72	0.0997
MobileNetV3	0.928	39.72	28.22	1577.53	3.67	0.0475
EfficientNet-B0	4.009	38.97	25.20	1518.69	15.58	0.1180
ResNet-18	11.177	45.49	33.76	2068.93	42.71	0.0791

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (6)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (7)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (8)$$

Here y_i and $\hat{y}_i$ denote the ground-truth and predicted irradiance values, respectively. In addition to forecasting accuracy, deployment-oriented metrics including parameter reduction, sparsity percentage, checkpoint size, compressed checkpoint size, and inference latency per image are also measured to evaluate model efficiency after compression.

3.2 Baseline Performance

The forecasting performance of the evaluated CNN architectures is presented in Table 1. All models were trained using infrared sky images with a spatial resolution of 60×80 pixels. MobileNetV2 achieved the lowest RMSE of 37.11 and the lowest MSE of 1376.78, indicating the strongest overall forecasting performance among the evaluated models. EfficientNet-B0 obtained the lowest MAE of 25.20, while its RMSE remained close to MobileNetV2. MobileNetV3 also achieved competitive performance, with an RMSE of 39.72 and a smaller checkpoint size of 3.67 MB. ResNet-18, despite having the largest checkpoint size of 42.71 MB, did not provide the best forecasting accuracy. On the contrary, EfficientNet-B0 is smaller than ResNet-18 (15.58 MB vs. 42.71 MB) yet slower (0.1180 ms/img vs. 0.0791 ms/img), while MobileNetV3 is the fastest of the non-TinyCNN models at 0.0475 ms/img. TinyCNN required only 0.06 MB of storage and 0.0200 ms per image, making it the smallest and fastest model, but its prediction error was noticeably higher than the other architectures. Overall, these results show that larger model size does not necessarily lead to better forecasting performance for low-resolution infrared sky images. MobileNetV2 provides the best accuracy, while MobileNetV3 offers a strong balance between accuracy, storage, and latency efficiency.

3.3 Structured Pruning

Structured pruning was applied to all baseline models using the torch-pruning framework at pruning levels of 20%, 40%, 60%, and 80%. Following pruning, each model was fine-tuned for three epochs and evaluated using RMSE, parameter reduction, model size, and inference latency. For comparison, the most compressed configuration below the 10% RMSE-increase threshold is reported; when no configuration met the threshold, the one with the smallest RMSE increase is shown instead. Table 2 shows that structured pruning removes a substantial share of parameters while largely preserving accuracy, with parameter counts falling by roughly a third to a half and RMSE increasing between 2.0% (ResNet-18) and 22.2% (MobileNetV3); MobileNetV2 and MobileNetV3 exceed the 10% threshold even at the lowest pruning level tested. TinyCNN attains the largest

Table 2: Selected structured-pruning configurations. For architectures with no configuration satisfying the 10% relative RMSE-increase threshold, the least-degrading tested configuration is reported.

Model	Prune (%)	Param. Red. (%)	Base RMSE (W/m ²)	Final RMSE (W/m ²)	Size (MB)	Latency (ms/img)
EfficientNet	20	35.37	38.97	40.97	10.15	0.1010
MobileNetV2	20	35.46	37.10	42.62	5.69	0.0772
MobileNetV3	20	35.19	39.72	48.52	2.42	0.0818
TinyCNN	40	48.87	82.14	86.12	0.03	0.0274
ResNet-18	20	36.26	45.48	46.40	27.25	0.1003

parameter reduction (48.87%) at a modest accuracy cost. Among the full-size architectures, EfficientNet is most accurate but slowest, ResNet-18 is most robust in RMSE despite being the largest model and among the slowest, and MobileNetV3 yields the smallest model at the largest RMSE increase, making it size-aggressive rather than best-balanced.

Figure 2a shows that 80% pruning degrades all architectures sharply, pushing validation RMSE well above 130 (and above 290 for EfficientNet, MobileNetV2/V3, and ResNet-18), indicating this ratio exceeds the viable compression limit for the task. TinyCNN dominates latency, running roughly 3-10 $\times$ faster than the other architectures, but its best configuration (20% pruning, RMSE $\approx$ 82) cannot match the accuracy of larger models at low pruning ratios, reflecting a capacity-versus-speed trade-off. At 20% pruning, MobileNetV2 and ResNet-18 form the accuracy-efficiency Pareto frontier: MobileNetV2 attains the lowest RMSE ($\approx$ 45) at $\approx$ 0.075 ms, while ResNet-18 reaches comparable RMSE ($\approx$ 50) at $\approx$ 0.10 ms but with a substantially larger parameter footprint, making MobileNetV2 the more parameter-efficient choice for high-accuracy deployment.

3.4 Unstructured Pruning

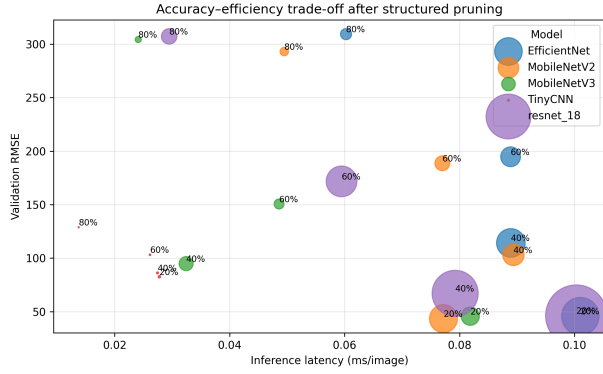
To evaluate the compressibility of the forecasting models, unstructured pruning was applied using L1-norm weight pruning on all convolutional and fully connected layers. Each model was pruned at 20%, 40%, 60%, and 80% sparsity levels and subsequently fine-tuned for 3 epochs. Since unstructured pruning preserves the original network architecture and does not reduce the number of stored parameters, compression effectiveness was evaluated using the size of gzip-compressed model checkpoints, in contrast to structured pruning, which reports the actual checkpoint size since parameters are physically removed. For each architecture, the best configuration was selected as the most compressed model that maintained an RMSE increase below 10% relative to the baseline model. Where no configuration met this threshold, the configuration with the smallest RMSE increase is reported instead.

Table 3 summarizes the selected configurations. EfficientNet, MobileNetV2, and ResNet-18 met the 10% RMSE threshold at approximately 20% sparsity, with RMSE increases of 3.6%, 3.0%, and 7.8%, respectively. TinyCNN tolerated the highest sparsity (60%) with only a 1.7% RMSE increase, but from a much higher baseline (82.14 W/m²). MobileNetV3 was the exception, showing a 31.7% RMSE increase (39.72 to 52.30 W/m²) at comparable sparsity, indicating higher sensitivity to unstructured weight removal. Excluding TinyCNN, MobileNetV2 anchors the accurate end with the lowest RMSE (38.23 W/m²). ResNet-18 offers the lowest latency among the full-size models (0.0887 ms/img) but the highest RMSE among them (49.02 W/m²), while MobileNetV3 combines the smallest footprint (2.90 MB) with the highest RMSE (52.30 W/m²) among the accurate models. Overall, unstructured pruning substantially reduces storage while largely preserving performance, but benefits are architecture-dependent and do not reliably translate into latency gains on dense hardware.

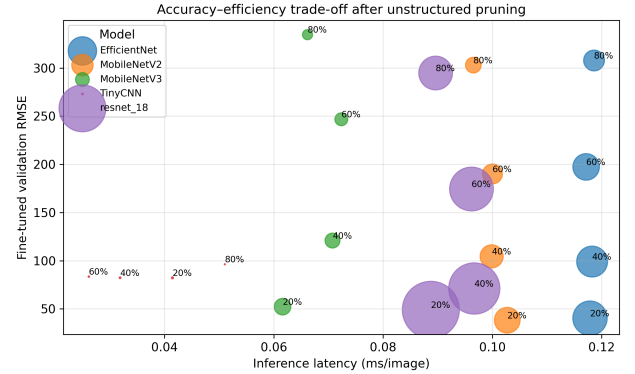
Figure 2b illustrates the resulting accuracy-efficiency trade-off. Across EfficientNet, MobileNetV2, and ResNet-18, latency remains essentially flat from 20% to 80% sparsity, confirming that unstructured sparsity yields no measurable hardware speedup. RMSE, in contrast, rises monotonically with pruning ratio across all architectures, reaching a 4-6 $\times$ increase over the 20% baseline by 80% sparsity. MobileNetV3 shows the sharpest trade-off: lowest latency at 20% pruning but the steepest accuracy collapse at 80%. ResNet-18 matches EfficientNet’s accuracy at every pruning level while sustaining roughly 25% lower latency, indicating a more favorable accuracy-latency profile despite its larger parameter count.

Table 3: Summary of unstructured pruning configurations selected using a 10% relative RMSE-increase comparison threshold. When no configuration met the threshold, the configuration with the smallest RMSE increase is reported.

Model	Prune (%)	Sparsity (%)	Base RMSE (W/m ²)	Final RMSE (W/m ²)	Size (MB)	Latency (ms/img)
EfficientNet	20	19.74	38.97	40.37	12.41	0.1178
MobileNetV2	20	19.69	37.10	38.23	6.94	0.1027
MobileNetV3	20	19.68	39.72	52.30	2.90	0.0616
TinyCNN	60	59.80	82.14	83.50	0.03	0.0261
ResNet-18	20	19.98	45.48	49.02	34.18	0.0887



(a) Accuracy–efficiency trade-off after structured pruning. Bubble size is proportional to model size.



(b) Accuracy–efficiency trade-off after unstructured pruning. Bubble size is proportional to the compressed checkpoint size.

Figure 2: Accuracy–efficiency trade-offs obtained using structured and unstructured pruning. Lower RMSE and lower inference latency indicate better forecasting efficiency, while bubble size represents the storage footprint of each pruned model.

4 Discussion and Conclusion

These results show that effective solar irradiance forecasting from low-resolution infrared sky images can be achieved using relatively lightweight CNN architectures: MobileNetV2 achieved the lowest baseline error despite being smaller than ResNet-18, and this relationship inverts under compression, where ResNet-18 tolerates structured pruning best while MobileNetV2 tolerates unstructured pruning best among the full-size architectures. MobileNetV3 is the consistent exception, degrading sharply under both strategies despite its compact baseline footprint. Under structured pruning at low pruning levels, MobileNetV2 reaches lower error than ResNet-18 at lower latency, despite ResNet-18’s greater robustness at higher pruning levels. Under unstructured pruning, latency stayed flat across all sparsity levels for every architecture even as error rose steadily, showing the compression gains were confined to storage rather than inference speed. Both strategies collapsed at the highest pruning ratio tested, with error degrading sharply across most architectures. Future work will explore knowledge distillation and mixed pruning schedules alongside quantization to test whether this compression ceiling can be extended.

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References

- [Alnajadah et al., 2025] Alnajadah, A., Bremer, C., Butler, S., Li, F., and TamizhMani, G. (2025). Intra-hour solar irradiance forecasting using sky cameras for pv plants. *Sustainable Energy Technologies and Assessments*, 84:104743.
- [Barhmi et al., 2026] Barhmi, K., Golroodbari, S. M., Knap, W., and Van Sark, W. (2026). Real-time solar irradiance forecasting for grid integration using all-sky imagery and multi-stage ai with kalman filter optimization. *Renewable Energy*, 259:125117.
- [Diagne et al., 2013] Diagne, M., David, M., Lauret, P., Boland, J., and Schmutz, N. (2013). Review of solar irradiance forecasting methods and a proposition for small-scale insular grids. *Renewable and Sustainable Energy Reviews*, 27:65–76.
- [Fang et al., 2023] Fang, G., Ma, X., Song, M., Mi, M. B., and Wang, X. (2023). Depgraph: Towards any structural pruning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16091–16101.
- [Guo, 2018] Guo, Y. (2018). A survey on methods and theories of quantized neural networks. *arXiv preprint arXiv:1808.04752*.
- [He and Xiao, 2023] He, Y. and Xiao, L. (2023). Structured pruning for deep convolutional neural networks: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 46(5):2900–2919.
- [Klotz and Nayar, 2026] Klotz, J. and Nayar, S. K. (2026). Forecasting solar energy using a single image. *Solar Energy*, 313:114540.
- [Menghani, 2023] Menghani, G. (2023). Efficient deep learning: A survey on making deep learning models smaller, faster, and better. *ACM Computing Surveys*, 55(12):1–37.
- [Nie et al., 2024] Nie, Y., Li, X., Paletta, Q., Aragon, M., Scott, A., and Brandt, A. (2024). Open-source sky image datasets for solar forecasting with deep learning: A comprehensive survey. *Renewable and Sustainable Energy Reviews*, 189:113977.
- [Nijhum et al., 2024] Nijhum, I. R., Kaloni, D., Kenny, P., and Dev, S. (2024). Enhancing intra-hour solar irradiance estimation through knowledge distillation and infrared sky images. In *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, pages 7207–7211.
- [Nijhum et al., 2026] Nijhum, I. R., Scanlon, C., and Dev, S. (2026). Design and evaluation of a dual-spectrum low-cost sky imager for enhanced solar prediction. *Measurement: Energy*, page 100103.
- [Paletta et al., 2023] Paletta, Q., Terrén-Serrano, G., Nie, Y., Li, B., Bieker, J., Zhang, W., Dubus, L., Dev, S., and Feng, C. (2023). Advances in solar forecasting: Computer vision with deep learning. *Advances in Applied Energy*, 11:100150.
- [Terrén-Serrano et al., 2021] Terrén-Serrano, G., Bashir, A., Estrada, T., and Martínez-Ramón, M. (2021). Girasol, a sky imaging and global solar irradiance dataset. *Data in Brief*, 35:106914.
- [Wu et al., 2020] Wu, H., Judd, P., Zhang, X., Isaev, M., and Micikevicius, P. (2020). Integer quantization for deep learning inference: Principles and empirical evaluation. *arXiv preprint arXiv:2004.09602*.
- [Zhu et al., 2025] Zhu, K., Hu, F., Ding, Y., Zhou, W., and Wang, R. (2025). A comprehensive review of network pruning based on pruning granularity and pruning time perspectives. *Neurocomputing*, 626:129382.